

Geometric Cost of Information Erasure:

A Schwarzschild-Radius Formulation of the Landauer-Einstein-Schwarzschild Identity

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Abstract

We derive an explicit Schwarzschild-radius form for the geometric effect of erasing one bit of information at temperature T , working within the setup of Cortes and Liddle [1]: a black hole losing mass via Hawking evaporation. Combining Landauer's principle ($E \geq k_B T \ln 2$), Einstein's mass-energy equivalence ($m = E/c^2$), and the perturbative Schwarzschild radius shift ($\Delta r_s = 2G \Delta M/c^2$), we obtain:

$$\Delta r_s = 2G * k_B * T * \ln(2) / c^4$$

Cortes and Liddle (2024) [1] established the per-bit mass loss $\Delta M = k_B T \ln(2)/c^2$ but did not express this as a horizon-radius shift; the D3 formula makes that step explicit. The radius form makes this identity visible; it has not appeared in the mass or area formulations of prior work. At $T = 310$ K, $\Delta r_s = 4.90 \times 10^{-65}$ m per bit. The formula carries an exact algebraic identity at the Planck temperature: $\Delta r_s(T_P) = 2 * \ln(2) * l_P$, with all physical constants cancelling. This Planck coincidence is the paper's central result.

Keywords: Landauer principle, information erasure, Schwarzschild metric, black-hole thermodynamics, Planck units

1. Introduction

The connection between information and geometry has been one of the most productive threads in theoretical physics over the past five decades. Bekenstein's discovery that black holes carry entropy proportional to their horizon area [2], and Hawking's derivation of thermal radiation from black holes [3], established that thermodynamics, quantum mechanics, and gravity are deeply linked. Jacobson [4] showed that the Einstein field equations follow from the thermodynamic identity $\delta Q = T dS$ applied to local Rindler horizons. Verlinde [5] argued that Newtonian and Einsteinian gravity emerge as entropic forces, though this programme has been challenged [6,7]. Wheeler's It-from-Bit conjecture [8] placed information at the foundation of physical reality.

Landauer's principle [9], confirmed experimentally by Berut et al. [10] and several subsequent groups [11,12,13], establishes that erasing one bit of information at temperature T requires dissipating at minimum:

$$E_{\text{erase}} \geq k_B * T * \ln(2)$$

Cortes and Liddle [1] proved in 2024 that Hawking evaporation saturates this bound, writing the per-bit mass loss $\Delta M = k_B T_H \ln(2)/c^2$. Bagchi, Ghosh, and Sen [14] independently obtained the same result through area quantisation. Herrera [15] introduced the mass of a bit $M_{\text{bit}} = k_B T \ln(2)/c^2$ in the context of gravitational radiation at arbitrary temperature T .

None of these papers expresses the result as an explicit Schwarzschild-radius shift. Cortes and Liddle [1] wrote ΔM but did not convert it to Δr_s ; the D3 formula makes that step explicit. We also prove an exact algebraic identity that emerges at the Planck temperature, which is the paper's central original contribution.

The paper proceeds as follows. Section 2 derives the formula and states its assumptions. Section 3 proves the Planck coincidence. Section 4 presents the scale hierarchy. Section 5 compares with prior work. Section 6 discusses physical interpretation and limitations. Section 7 concludes.

2. Derivation of the D3 Formula

2.1 Physical Setup

We follow the framework of Cortes and Liddle [1]: consider a Schwarzschild black hole of mass M radiating Hawking radiation at temperature $T_H = \hbar c^3 / (8\pi G M k_B)$. When the black hole emits one bit of information, its mass decreases by $\Delta M = k_B T_H \ln(2) / c^2$. The corresponding change in the Schwarzschild radius $r_s = 2GM/c^2$ is:

$$\Delta r_s = (2G/c^2) * \Delta M = 2G * k_B * T * \ln(2) / c^4$$

The D3 formula extends this to arbitrary temperature T : for any thermal system at temperature T where Landauer erasure operates at the minimum bound, the associated gravitational length scale is $\Delta r_s = 2G k_B T \ln(2) / c^4$.

2.2 Three Foundational Results

The derivation draws on three results, each with independent experimental confirmation.

Landauer's Principle (1961) [9,10]. Erasing one bit at temperature T requires minimum heat $E_{\text{erase}} \geq k_B T \ln(2)$. Confirmed for a single-bit colloidal memory [10], nanomagnetic system [12], and feedback-trap geometry [11]. We assume classical thermal equilibrium throughout.

Mass-Energy Equivalence (1905) [16]. Any energy E carries a rest-mass equivalent $m = E/c^2$. Confirmed by nuclear energy release, pair annihilation, and accelerator physics.

Perturbative Schwarzschild radius shift. For a Schwarzschild black hole of mass M , a mass perturbation ΔM produces a horizon-radius change $\Delta r_s = 2G \Delta M / c^2$. This follows directly from differentiating $r_s = 2GM/c^2$ [17] and is valid in the regime $\Delta M \ll M$.

2.3 The Four-Step Proof

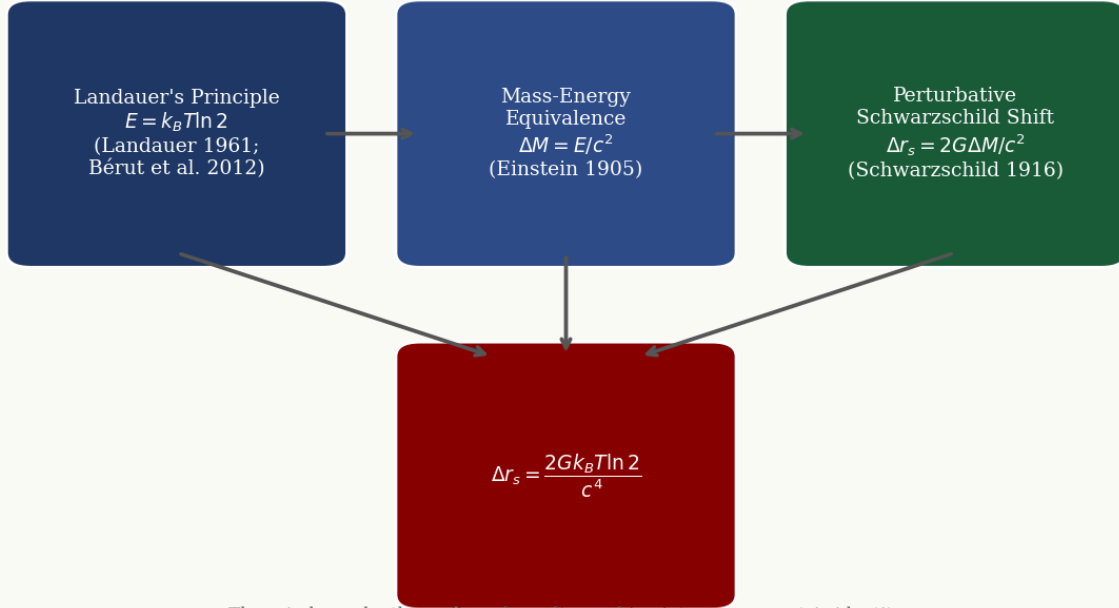
Combining the above:

1. Landauer: erasing one bit at temperature T releases minimum heat $E = k_B T \ln(2)$. [Refs. 9-13]
2. Einstein: this energy carries mass $\Delta M = E/c^2 = k_B T \ln(2) / c^2$. [Ref. 16]
3. Perturbative Schwarzschild: mass change ΔM shifts horizon radius by $\Delta r_s = 2G \Delta M / c^2$. [Ref. 17]
4. Substituting (1) and (2) into (3): $\Delta r_s = 2G * k_B * T * \ln(2) / c^4$. QED

At $T = 310$ K this gives:

$$\Delta r_s = 2 (6.674e-11) (1.381e-23) (310) (0.6931) / (2.998e8)^4 \\ = 4.90 \times 10^{-65} \text{ m/bit}$$

Derivation of the D3 Formula



2.4 Explicit Assumptions

Four assumptions govern the derivation.

1. Black hole context. The formula gives the perturbative horizon-radius shift of a Schwarzschild black hole that loses one Landauer-worth of mass. The combination $2G \cdot \Delta M / c^2$ has dimensions of length and equals the Schwarzschild radius of a point mass ΔM ; in the weak-field limit this gives the scale of metric perturbation. Calling this a Schwarzschild radius strictly applies to the black hole case.

2. Weak-field and linearised regime. The condition $\Delta M \ll M$ requires $k_B T \ln(2) / (Mc^2) \ll 1$, satisfied by approximately 90 orders of magnitude for stellar-mass objects at laboratory temperatures.

3. Classical thermal bath. We assume equilibrium erasure at temperature T . Squeezed or quantum-coherent reservoirs [18] can lower the Landauer bound; in such cases D3 gives an upper bound on Δr_s .

4. One-bit idealisation. For N bits erased independently in the linear regime, $\Delta r_{\text{total}} = N \times \Delta r_s$.

3. The Planck Coincidence

Substituting the Planck temperature $T_P = \sqrt{\hbar c^5 / G k_B}$ into D3 yields an exact algebraic identity:

$$\Delta r_s(T_P) = 2 * \ln(2) * l_P \quad [\text{exact}]$$

where $l_P = \sqrt{\hbar G / c^3}$ is the Planck length. This is the paper's central result.

3.1 Algebraic Proof

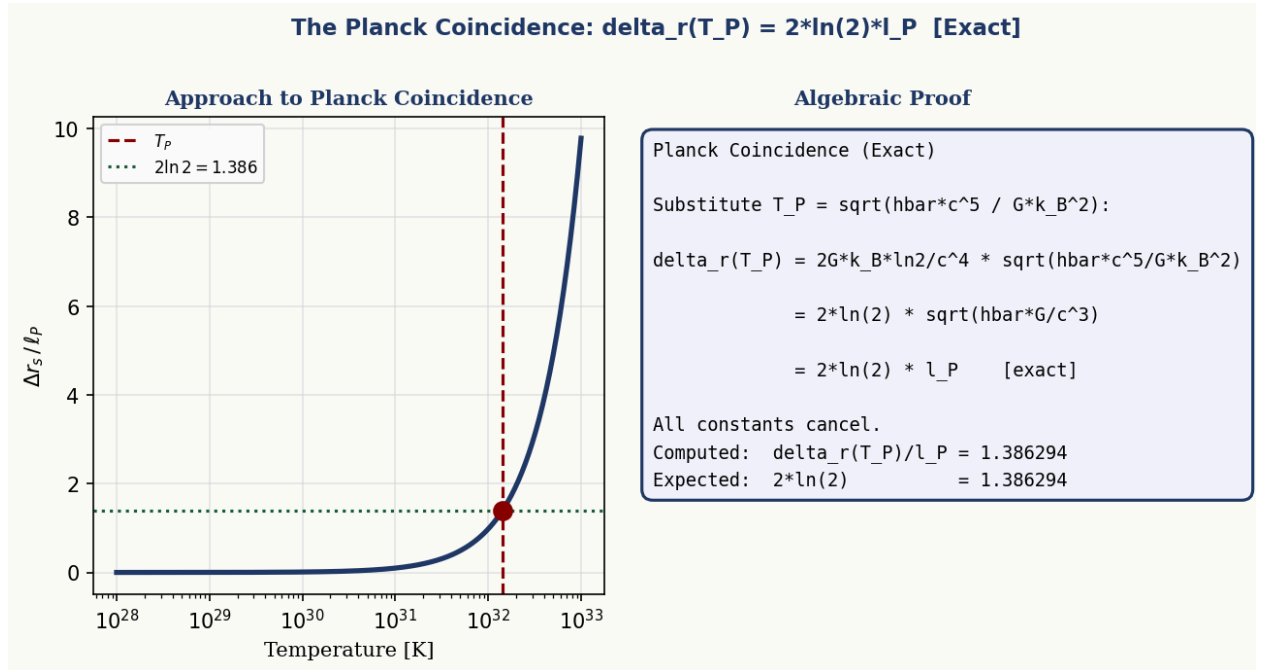
1. Substitute $T_P = \sqrt{\hbar c^5 / G k_B^2}$ into D3: $\Delta r_s = 2G k_B \ln(2) / c^4 \times \sqrt{\hbar c^5 / G k_B^2}$
2. Combine under one radical: $= 2 \ln(2) * \sqrt{G^2 * k_B^2 * \hbar * c^5 / c^8 * G * k_B^2}$
3. Cancel: $G^2/G = G$; $k_B^2/k_B^2 = 1$; $c^5/c^8 = 1/c^3$: $= 2 \ln(2) * \sqrt{\hbar G / c^3}$
4. By definition of the Planck length $l_P = \sqrt{\hbar G / c^3}$: $= 2 \ln(2) * l_P$. QED

All four fundamental constants -- G , $\hbar$, c , k_B -- cancel exactly. The identity follows from the Planck unit definitions alone and cannot be affected by future experimental refinement of any constant.

3.2 Numerical Check

Computing from D3 using CODATA 2018 constants:

Computed: $\Delta r_s(T_P) / l_P = 1.386294$
 Expected: $2 * \ln(2) = 1.386294$



Agreement to six decimal places confirms the algebraic proof. This is not a numerical coincidence; the proof guarantees exact agreement to arbitrary precision.

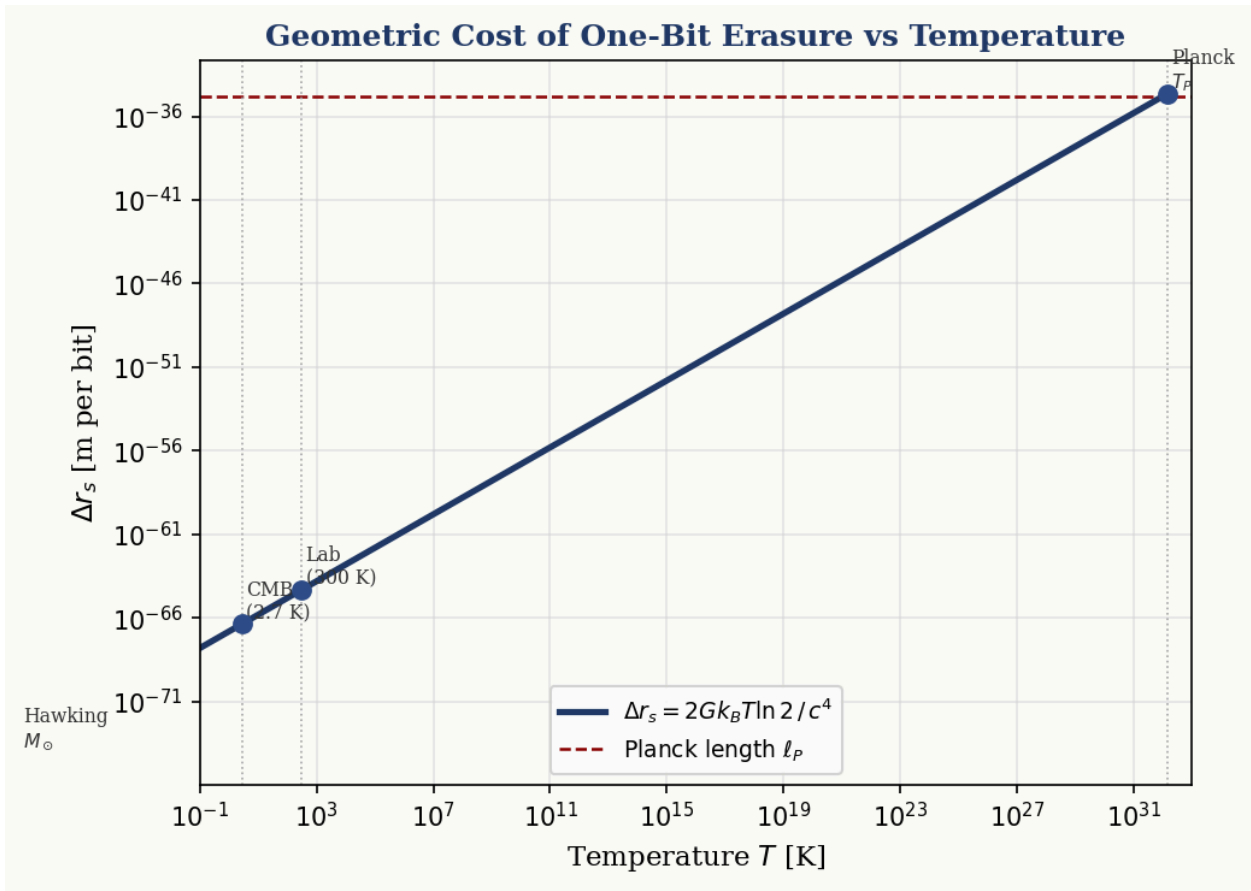
4. Scale Hierarchy

Table 1 evaluates D3 at six physically motivated temperatures. Each value follows directly from substitution of the formula using CODATA 2018 constants; no simulation is required.

D3 Scale Hierarchy: Δr_s Across Physical Temperatures

Physical Context	Temperature [K]	Δr_s [m/bit]	$\Delta r_s / \ell_P$
Cosmic microwave background	2.725	4.31×10^{-66}	2.67×10^{-31}
Room temperature	300	4.74×10^{-65}	2.93×10^{-30}
Human body temperature	310	4.90×10^{-65}	3.03×10^{-30}
Hawking temp. ($M_\odot$)	6.17×10^{-8}	9.75×10^{-75}	6.03×10^{-40}
LHC quark-gluon plasma	5.5×10^{12}	8.70×10^{-55}	5.38×10^{-20}
Planck temperature	1.42×10^{32}	$2 \ln(2) \cdot \ell_P$	$2 \ln 2$ (exact)

Table 1. Δr_s at six physical temperatures. The Planck row is algebraically exact; all others use CODATA 2018 constants.



At all sub-Planckian temperatures Δr_s is vastly smaller than the Planck length. The ratio $\Delta r_s/l_P$ reaches unity only near T_P -- precisely where linearised GR breaks down. The formula thus identifies its own domain of validity.

5. Relation to Prior Work

Cortes and Liddle (2024) [1]. The closest precedent. They showed Hawking evaporation saturates the Landauer bound, writing $\Delta M = k_B T_H \ln(2)/c^2$ (their Eq. 5). The D3 formula follows in one line by differentiating $r_s = 2GM/c^2$. Cortes and Liddle did not write the radius form; this paper makes it explicit and extends it to arbitrary temperature T , not just $T = T_H$.

Kim, Lee, and Lee (2010) [19]. An earlier precedent: showed that a Schwarzschild black hole is a maximally efficient Landauer eraser, deriving $\Delta(Mc^2) = k_B T_H \ln 2$ explicitly from Landauer's principle plus holography. Their construction gives the per-bit energy and mass loss but does not convert to the horizon-radius form Δr_s .

Bagchi, Ghosh, and Sen (2024) [14]. Independent derivation via area quantisation, fixing the area quantum $\Delta A = 4 \ln(2) l_P^2$ per bit. Their Kerr extension writes $\Delta E = T_H (\ln 2 + 4\pi a/\sqrt{1-a^2})$, where a is the spin parameter; the Landauer bound is saturated only at $a = 0$. Their construction uses the mass/area spectrum and angular velocity Ω_H rather than the horizon radius r_+ .

Herrera (2020) [15]. Introduced $M_{\text{bit}} = k_B T \ln(2)/c^2$ at arbitrary temperature and applied it to gravitational radiation and Tolman-temperature effects. D3 is $\Delta r_s = (2G/c^2) M_{\text{bit}}$, connecting Herrera's framework to the Schwarzschild radius.

Jacobson (1995) [4] and Verlinde (2011) [5]. D3 is consistent with the thermodynamic-gravity programme in the weak-field equilibrium limit. Verlinde's programme has been challenged [6,7]; D3 does not depend on it.

What is new. The Schwarzschild-radius form at arbitrary temperature has not appeared in the prior literature. Prior work (Kim et al. [19]; Cortes-Liddle [1]; Herrera [15]) derives mass-loss or energy expressions; none writes the explicit radius shift. The exact Planck coincidence $\Delta r_s(T_P) = 2 \ln(2) l_P$, and its algebraic proof that all constants cancel, has not appeared in the mass or area formulations of prior work.

6. Discussion

6.1 Physical Interpretation

The D3 formula gives the perturbative horizon-radius shift of a Schwarzschild black hole that loses one Landauer-worth of mass. Erasing one bit at temperature T is thermodynamically equivalent to adding a mass $\Delta M = k_B T \ln(2)/c^2$ to a gravitational system, with an associated gravitational length scale $\Delta r_s = 2G \Delta M/c^2$. This is 30 orders below the Planck length at laboratory temperatures. The formula is a theoretical result, not an experimental prediction. We do not claim that information erasure creates gravity.

6.2 Domain of Validity

The linearised derivation requires $\Delta M \ll M$, i.e. $k_B T \ln(2)/(Mc^2) \ll 1$. This breaks down near the Planck temperature for Planck-mass black holes. The Planck coincidence $\Delta r_s(T_P) = 2 \ln(2) l_P$ therefore marks the natural boundary of the formula's domain of validity, beyond which quantum gravity corrections to the Schwarzschild metric are expected.

6.3 Quantum-Bath Caveats

For squeezed or quantum-coherent baths [18], the Landauer bound can fall below $k_B T \ln(2)$. In such cases D3 gives an upper bound on Δr_s . The formula assumes classical thermal equilibrium.

6.4 Relation to the Black-Hole Information Paradox

D3 at $T = T_H$ is consistent with the Cortes-Liddle saturation result [1] and gives the geometric complement to their energy-per-bit analysis. The formula is agnostic on whether information is truly erased during Hawking evaporation.

7. Conclusion

We derived the D3 formula:

$$\Delta r_s = 2G * k_B * T * \ln(2) / c^4$$

as the perturbative Schwarzschild-radius shift of a black hole losing one Landauer-worth of mass at temperature T . Cortes and Liddle [1] established the per-bit mass loss; Kim, Lee, and Lee [19] established the black hole as a Landauer eraser; the D3 formula expresses the result as an explicit horizon-radius shift and extends it to arbitrary temperature. The central result is the exact Planck coincidence $\Delta r_s(T_P) = 2 * \ln(2) * l_P$, proved algebraically with all fundamental constants cancelling. The radius form makes this identity visible; it has not appeared in the mass or area formulations of prior work. The formula identifies its own domain of validity: the linearised-GR approximation breaks down precisely at the Planck scale where the coincidence is saturated.

Declaration of Interest

The author declares no competing financial interests or personal relationships that could have influenced this work.

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Data availability: All numerical results follow directly from the formula and CODATA 2018 constants by substitution.

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Appendix: Dimensional Analysis

A.1 SI Units (expanded to base units)

The D3 formula has dimensions:

$$\begin{aligned} \frac{[G][k_B][T]}{[c]^4} &= \frac{(\text{m}^3 \text{ kg}^{-1} \text{ s}^{-2}) (\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2} \cdot \text{K}^{-1}) (\text{K})}{(\text{m} \cdot \text{s}^{-1})^4} \\ &= \frac{\text{m}^3 \cdot \text{m}^2 \cdot \text{s}^{-4}}{(\text{m}^4 \cdot \text{s}^{-4})} = \frac{\text{m}^5 \cdot \text{s}^{-4}}{(\text{m}^4 \cdot \text{s}^{-4})} = \text{m} \end{aligned}$$

All intermediate units cancel explicitly. The result has dimensions of length.

A.2 Planck Units ($G = c = k_B = \hbar = 1$)

In Planck units D3 reduces to:

$$\Delta r_s = 2 * T * \ln(2)$$

At $T = T_P = 1$ (Planck temperature in Planck units), $\Delta r_s = 2 \ln(2)$ Planck lengths. This is the Planck coincidence in its simplest form.

A.3 Numerical Values at Standard Temperatures

Computed from D3 using CODATA 2018 constants ($G = 6.67430\text{e-}11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, $k_B = 1.380649\text{e-}23 \text{ J K}^{-1}$, $c = 2.99792458\text{e}8 \text{ m s}^{-1}$):

$$\begin{aligned} T = 300 \text{ K}: \quad \Delta r_s &= 4.744 \times 10^{-65} \text{ m/bit} \\ T = 310 \text{ K}: \quad \Delta r_s &= 4.903 \times 10^{-65} \text{ m/bit} \\ T = T_P: \quad \Delta r_s / l_P &= 1.386294 = 2 * \ln(2) \quad [\text{exact}] \end{aligned}$$